

RETAIL DEMAND & INVENTORY PLANNING

Independent supply chain analytics project | Microsoft Excel

PROJECT OBJECTIVE

Determine how a retailer should forecast weekly demand and set inventory reorder points for its highest-revenue products while balancing product availability and excess inventory.

541,909	23,243	20	52	17.8%
source records	transactions analyzed	SKUs modeled	complete weeks	error improvement

METHOD

- Cleaned cancellations, returns, zero-price lines, and non-inventory service codes.
- Aggregated 20 high-revenue physical SKUs into 52 complete demand weeks.
- Backtested one-week naive, four-week moving-average, and eight-week moving-average forecasts over an eight-week holdout period.
- Built ABC segmentation, safety stock, and reorder-point policies using a two-week lead time and 95% service target.

KEY FINDINGS

- Four-week moving average: **44.9% portfolio WMAPE**.
- Forecast error improved **17.8%** versus the one-week naive baseline.
- A-class products warrant weekly review because they represent most modeled revenue.
- Highly variable products require exception management instead of one blanket inventory rule.

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Tools and concepts: Excel formulas, time-series aggregation, forecast backtesting, WMAPE, ABC inventory classification, service levels, safety stock, reorder points, dashboard design, and management recommendations.

Data source: UCI Machine Learning Repository, Online Retail dataset. <https://archive.ics.uci.edu/dataset/352/online+retail> | CC BY 4.0